

MOSES MAINA ONGWARE

Systems-Driven Software Engineer

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SUMMARY

Systems-driven software engineer and Computer Science graduate experienced in building reliable custom software, secure APIs, and production-ready full-stack systems. Strong in software architecture, backend engineering, database design, system integrations, performance optimization, and automated testing. Uses systems thinking and AI-assisted workflows to clarify requirements, evaluate tradeoffs, and deliver maintainable software that solves complex operational problems.

PROFESSIONAL EXPERIENCE

IT Support Trainee | *Rural Electrification & Renewable Energy Corporation (REREC)* May 2024 – Aug 2024

- Supported SAP S/4HANA onboarding and troubleshooting for enterprise users.
- Managed Jira service requests covering hardware maintenance and IT operations.
- Monitored server-room infrastructure and assisted with device and network issue resolution.

PROFESSIONAL PROJECTS

WhatsApp Product Verification (WPV) | [WPV Demo](#)

- Developed a Node.js/TypeScript anti-counterfeiting platform supporting large-scale WhatsApp product verification, with an operations dashboard for code management, investigations, analytics, and reporting.
- Prevented duplicate activations and race-condition fraud using PostgreSQL transactions, row-level locking, automatic blocking, audit trails, JWT authentication, HMAC-SHA256 webhook validation, and rate limiting.
- Achieved sub-24ms p95 response times with zero failed requests during load testing at 200 concurrent users through indexing, query optimization, and connection pooling.

Rental Management System | [Admin Demo](#) | [Landlord Demo](#) | [Tenant Demo](#) | [Caretaker Demo](#)

- Architected a multi-tenant PropertyTech platform comprising four independently deployable React/TypeScript portals and a role-scoped Node.js/Express API for administrators, landlords, tenants, and caretakers.
- Engineered race-safe M-Pesa rent reconciliation using idempotent callbacks, Serializable transactions, PostgreSQL advisory locks, unique receipt constraints, and missed-callback recovery to prevent duplicate credits and double-billing.
- Built rent-cycle accounting, instalment plans, maintenance workflows, automated reminders, analytics, and asynchronous PDF/Excel reporting using PostgreSQL, Prisma, Redis, and BullMQ.
- Secured the platform with RS256 authentication, refresh-token rotation and revocation, RBAC, rate limiting, account-status enforcement, and audit logging.
- Maintained 263 passing backend tests across 67 suites, covering authentication, authorization, payment reconciliation, rate limiting, exports, and role-specific workflows.

TaskFlow | [TaskFlow Repository](#) | [Try TaskFlow](#)

- Built a TypeScript task-management platform using Next.js 14, Express, MongoDB/Mongoose, and JWT authentication, covering protected workspaces, project dashboards, Kanban boards, email verification, and password reset flows.
- Designed a project-board-task data model with owner/member access control, ordered board columns, status-driven task movement, default board provisioning, cascading project cleanup, Zod request validation, and tested ObjectId access utilities.
- Optimized data access and UI freshness with MongoDB aggregation pipelines, selective population, lean queries, server-side pagination, compound/text indexes, Redis response caching, and centralized React Query invalidation across dashboard, project, board, and task views.

EDUCATION

Bachelor of Science in Computer Science | **Chuka University** Graduated Nov 2025

Relevant Coursework: Software Engineering, User Interface Design, Distributed Systems, Decision Support Systems, Component Based Architecture, Software Project Management, Artificial Intelligence, Data Structures.

TECHNICAL STACK

Languages: TypeScript, JavaScript, Python, SQL, Dart, PHP

Frontend: React, Next.js, Vue.js, Flutter, Tailwind CSS, Material UI, shadcn/ui, TanStack Query, React Router, Vite

Backend & APIs: Node.js, Express.js, FastAPI, Laravel, REST API Development, Webhooks, Background Jobs, JWT Authentication, RBAC, BullMQ

Databases & Data: PostgreSQL, MongoDB, Redis, Prisma ORM, Database Design, Transactions, Query Optimization, Caching

Integrations: M-Pesa Daraja API, Meta WhatsApp Cloud API, Cloudinary, Resend, OAuth

Tests & Quality: Jest, Supertest, Playwright, k6, Integration Testing, Load Testing, API Testing

DevOps & Observability: Docker, GitHub Actions, Linux, Git, GitHub, Vercel, Render, Prometheus, Winston, Pino, ngrok

Engineering & Competencies: Software Architecture, Systems Thinking, Custom Software Development, Backend Engineering, API Design, Security Engineering, Performance Optimization, Technical Problem Solving

CERTIFICATION

Docker Bootcamp for ML, GenAI and Agentic AI • Prompt Engineering with Amazon Q Developer • Cybersecurity & Emerging Technologies Awareness